

UTM

Universal Triage Model

universal approach for supporting industrial workflows

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Model Architecture

Model Pipeline

GPT-3.5 for data generation, Llama 3.2 baseline, RoBERTa fine-tuning

Input Processing

- Company background (industry, enterprise information system)
- Workflow description
- Components
- User report ticket

Output

- Affected component
- Severity level (Medium, High, Critical)



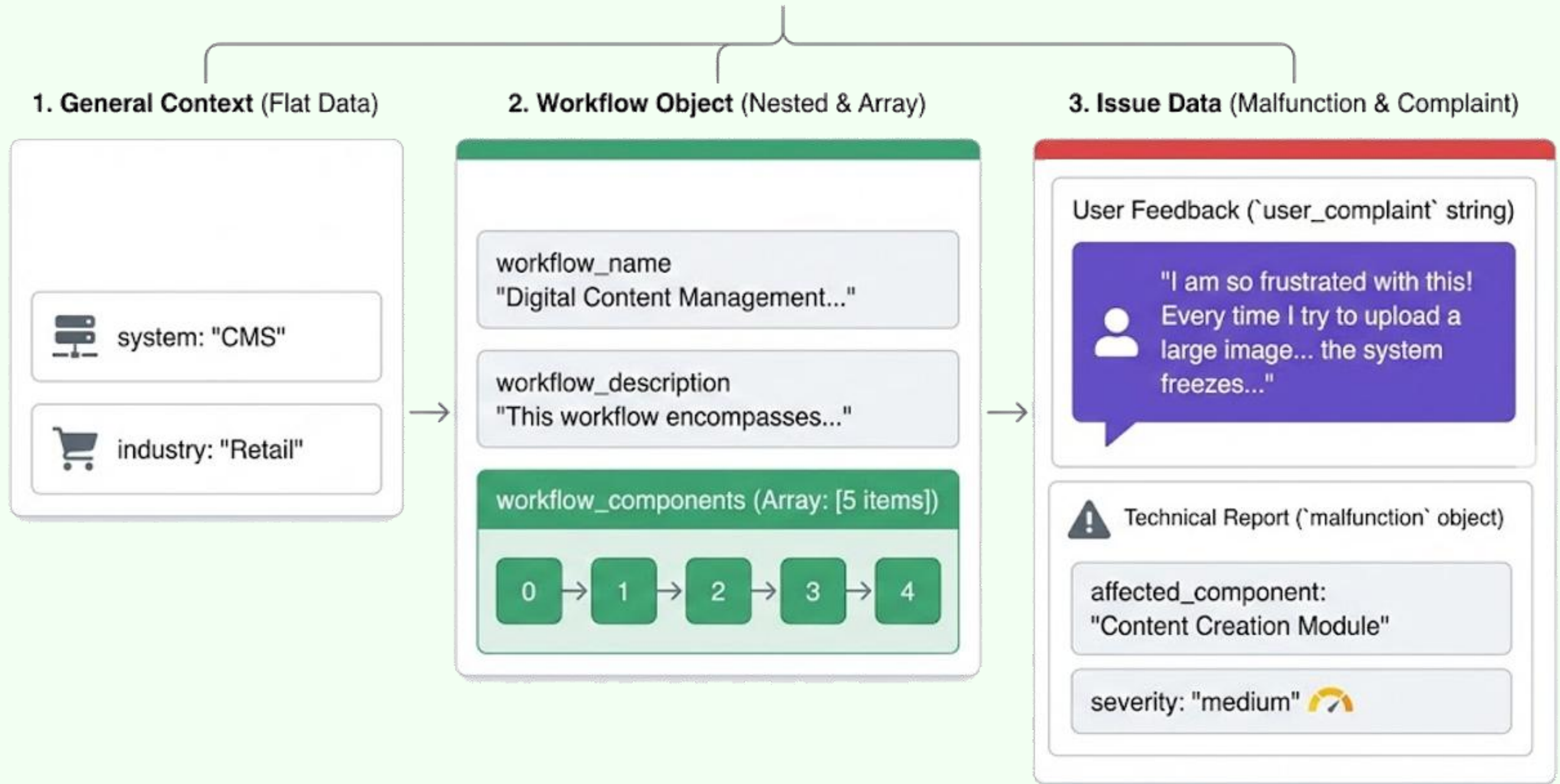
Component Metrics

Accuracy, Top-k ranking

Severity Metrics

F1 Score (macro), Accuracy, Weighted F1

DATASET STRUCTURE: The Dashboard View



Problem & Solution

Current Challenge

Manual ticket classification is costly and time-consuming

- High operational costs
- Slow response times

Our Innovation

Universal AI model adaptable across varied software systems

- Automated classification
- Multi-company deployment



Model Development & Data Strategy

Data Generation

~2220 training examples, ~400 test examples

Baseline Models

Llama 3.2 for zero-shot and few-shot testing

Fine-Tuned Models

Component Classification

- Binary relevance classifier & ranking model
- Metrics: F1, Accuracy, Recall, Precision

Severity Classification

- RoBERTa-base for Critical/High/Medium
- Metrics: F1 (macro & weighted), Accuracy, Recall, Precision

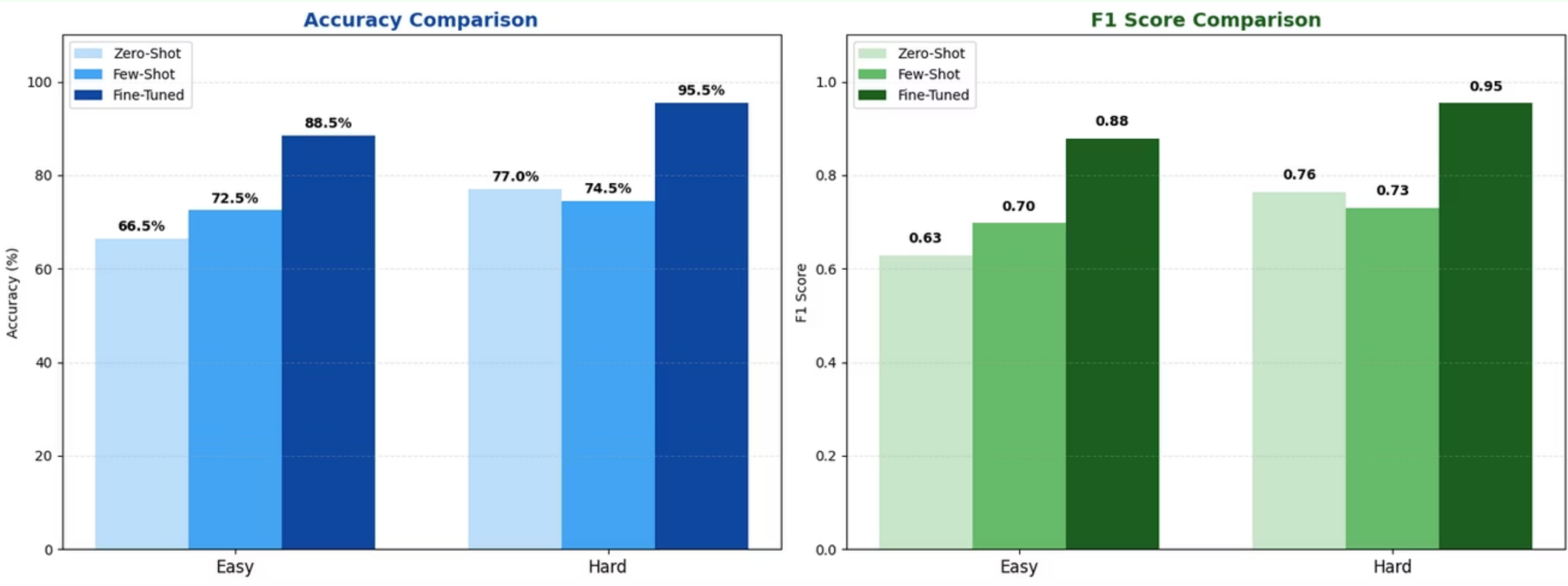
Refinement Efforts

- Increase data diversity & quantity
- Adjust metrics (eval loss for best model)
- Component classification formulated as a ranking problem

Category		Component Model	Severity Model
	Main Task	Binary Classification (Ranker Approach)	Multi-class Classification (3 severity levels)
	Data Preparation	Explosion: Splitting each issue into separate rows for every component	Contextual Prompt: Including Industry, System, and User Experience
	Number of Epochs	3	10 (with Early Stopping)
	Learning Rate (LR)	3e-5	1e-5
	Layer Freezing	None (Full training)	Freezing the first 4 layers
	Imbalance Management	WeightedTrainer (weights for Label 1)	WeightedTrainer (weights for Critical and High)
	Primary Metrics	Top-2 Accuracy and F1	Macro-F1 and Accuracy
	Regularization	Early Stopping, Weight Decay	Early Stopping, Weight Decay, Layer Freezing, Warmup

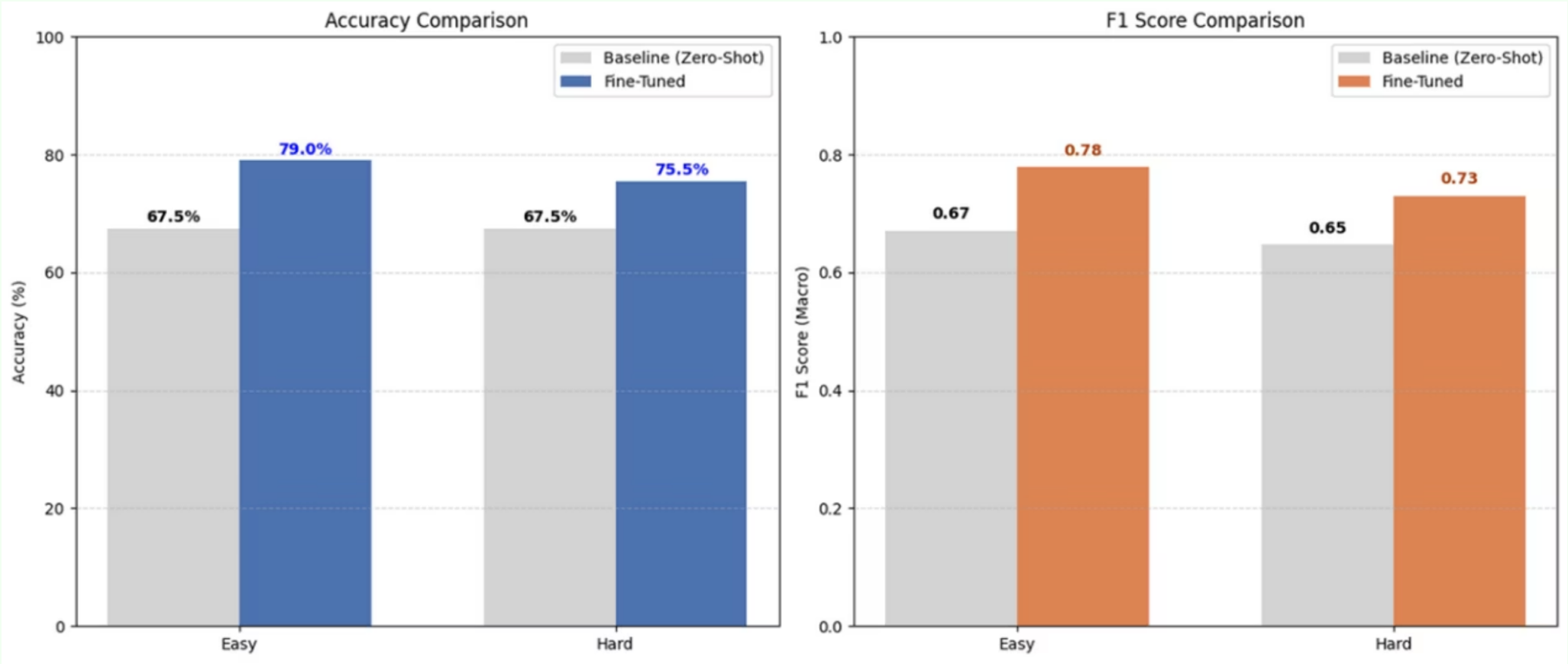
Results: Seen vs. Unseen Context

Component Classification Accuracy



Dataset	Zero-Shot	Few-Shot	Fine-Tuned	Gain
Easy	66.5%	72.5%	88.5%	+16.0%
Hard	76%	73%	95.5%	+19.5%

Severity Classification Performance



Dataset	Baseline Acc	Fine-Tuned Acc	Baseline F1	Fine-Tuned F1
Easy	67.5%	79%	0.6708	0.7783
Hard	67.5%	75.5%	0.6475	0.7303

Key Learnings & Future Directions

Achievement

Successfully trained universal model that predicts components and severity on unseen workflows

Critical Lessons

- Clear label definitions essential
- Testing on unseen data is crucial

Future Exploration

Real-World Data

Deploy with multiple companies, collect real tickets for enhanced training

Active Learning

User corrections as supervision signals for continuous model improvement

